1. Determine whether the sequence converges or diverges. If it converges, find the limit. If it diverges, explain why.

   a) $a_n = \dfrac{4n-3}{3n+4}$ b) $a_n = \sin\dfrac{n\pi}{2}$

2. Determine whether the sequence converges or diverges. If it converges, find the limit. If it diverges, explain why.

   a) $a_n = \arctan\left(\dfrac{2n}{2n+1}\right)$ b) $a_n = (1+3n)^{1/n}$

3. Determine whether the sequence converges or diverges. If it converges, find the limit. If it diverges, explain why.

   $a_n = \ln(\sqrt{2n^4+3}) - \ln(9+4n^2)$

The Squeeze Theorem for Sequences

If $\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} b_n$ and there exists an integer N such that $a_n \le c_n \le b_n$ for all $n \ge N$, then $\lim_{n\to\infty} c_n = L$.

4. Find the limit of $a_n = \dfrac{\tan^{-1} n}{n}$ using the squeeze theorem. [Draw a "picture". Think of the function $\tan^{-1}(x)$.]

5. Determine whether the sequence is increasing, decreasing or not monotonic. Is the sequence bounded? [Besides thinking of the formal definition of "bounded", think about what that words means in context.]

   a) $a_n = n(-1)^n$

   b) $a_n = \dfrac{n-2}{n+2}$

6. Suppose you know that a sequence $\{a_n\}$ is a decreasing sequence and all its terms lie between the numbers 5 and 8. Explain why the sequence has a limit. What can you say about the value of the limit?